

Review

Exploring Key Parameters in Adsorption for Effective Fluoride Removal: A Comprehensive Review and Engineering Implications

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Abstract: Water pollution, particularly from elevated fluoride ion (F^-) concentrations, is a significant challenge in many developing countries, particularly those relying on groundwater. The stable form of fluoride, F^- , poses health risks, leading to concerns about various diseases and harmful effects. Despite global efforts, high F^- concentrations ($>1.5 \text{ mg L}^{-1}$) persist in numerous countries, requiring effective and sustainable removal methods. Adsorption, known for its simplicity, cost-effectiveness, and efficiency, stands out as a promising technique for F^- removal from drinking water. Successful commercial implementation necessitates the optimization of separation conditions. This systematic literature review focuses on the adsorption process for F^- removal, exploring parameters such as temperature, adsorbent particle size, pH, adsorbent mass, and co-existing ions for efficient removal. Observations indicate that, despite the utilization of a diverse range of adsorbents, several limitations persist. These include low adsorption capacity, a sluggish adsorption rate, a restricted pH range, and high associated costs. The mechanistic understanding of adsorption and the ongoing development of novel adsorbents remain focal points for future research. Additionally, there is a need to explore alternative kinetic models grounded in statistical factors and give due consideration to thermodynamic studies.

Keywords: fluoride removal; water pollution; adsorption; groundwater; sustainable treatment



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1. Introduction

Water pollution is a major problem in developing countries, especially in regions where there is dependence on groundwater as a primary source of supply [1,2]. The availability of drinking water is gradually decreasing due to the deposition of organic and inorganic wastes in the aquatic environment. Almost 40% of the world's population is facing quantitative water scarcity. Moreover, water quality in rivers and groundwater has been affected by pollution due to waste and contaminants from cities, industries, and agriculture [2].

Groundwater springs constitute good quality water sources in urban areas as well as rural areas because they are relatively free of organic contaminants [3,4]. However, the chemical characteristics of groundwater play an important role in determining whether it is available for use [5,6]. The presence of certain inorganic ions, in low or high concentrations, in groundwater makes it unsuitable for consumption [3]. Thus, the excess of these ions

must be removed. These constituents, which may be present naturally or contaminating water sources due to anthropic activity, have become a major public health problem [2,7,8].

Fluoride ion (F^-) is the stable form of fluoride found in the environment and combines with other elements [9–11], and is the 13th most abundant element in the Earth's crust [9]. Fluoride is a naturally occurring mineral that is commonly added to public water supplies in order to help prevent tooth decay. However, in some cases, fluoride can be present in water sources at levels that are higher than what is considered safe for consumption. Fluoride is found in all natural waters in some concentration [9,10]; however, being one of the most abundant contaminants in water [12], it is most commonly found in groundwater [13].

Human health can be affected by insufficient or excessive presence of fluoride. In severe cases, it can also result in skeletal fluorosis, which can harm bones and joints and cause joint pain and stiffness. Dental caries can develop in concentrations between 0 and 0.5 mg L^{-1} , while dental fluorosis can appear in concentrations between 1.5 and 4 mg L^{-1} . In addition, there is an increased risk of skeletal fluorosis, paralysis, disability, or even death for higher concentrations of fluoride [14]. According to Craig et al. [1], most fluoride-related health problems occur in poor communities in the developing world, where F^- concentrations in groundwater are high and primary sources of drinking water are from hand-drilled or artesian wells.

Various diseases and harmful effects caused by F^- -contaminated water have been of great concern to society [2,15]. In some villages in China, it exceeds 8 mg L^{-1} , and in some areas of India, it was as high as 30 mg L^{-1} [16]. There were further reports in the literature that, in some countries, the concentration of F^- in drinking water can be higher than $30\text{--}35 \text{ mg L}^{-1}$ [7,17–20]. Several studies described the occurrence of high concentrations of F^- in drinking water, notably in countries in America, Africa, and Asia [21,22]. Thus, it is evident that the presence of F^- in water supplies is still a problem to be solved in many countries around the world [8]. In view of this, some kind of special treatment should be employed to remove the excess of this ion [4,6,23–27]. Then, monitoring and controlling the quantity of fluoride in public water supplies is crucial to preventing exposure to high fluoride levels. Additionally, it is crucial to make sure that the fluoride levels in private wells and other water supplies are periodically checked. Fluoride can be removed from water using treatment techniques if it is found in high concentrations [17,23,28–30].

Adsorption, as a medium for separating mixtures, has seen considerable advancement and is often touted as a simple and affordable means to eliminate excess fluoride from drinking water [12,20,22,31,32]. Despite its cost-effectiveness and extensive exploration for F^- removal, transitioning this method to commercial scale necessitates an adsorbent that meets the criteria of availability, quantity, and economic and environmental feasibility. With this perspective, research is fundamental for the search of new adsorbents and optimization of separation conditions.

Some review papers concerning the adsorption process for fluoride removal have been published [27,31,33–35]. However, none of them deeply investigated the parameters affecting the adsorption process. In this context, the aim of this review is to provide a systematic, critical, updated, and more in-depth analysis of the specific parameters to be considered for fluoride removal by adsorption, such as pH, adsorbent dosage, surface area, pore size of the adsorbent, temperature, and competing ions. This review is structured following the scope of the articles and is organized into three main topics, including general principles for fluoride adsorption, factors influencing the adsorption of this compound, fluoride adsorption capacity and engineering implications. The objectives and key results obtained by the authors of the selected articles were thoroughly analyzed and appropriately compared.

2. Methodology

For the critical review of the specific parameters to be considered for fluoride removal by adsorption, Scopus, ScienceDirect, and Web of Science were used as the main repositories to find published references on the topic. The search strategy included different combinations of the following key terms: “fluoride removal, water pollution, adsorption

parameters, groundwater, sustainable treatment". To filter the retrieved sources, only research articles from the last 15 years (2010–2024) that utilized an adsorption process for fluoride removal in aqueous phase were considered. Thus, all works where fluoride removal occurred through another treatment process, or where adsorption was used in gas treatment, were discarded. A bibliometric analysis was realized, employing various bibliometric ranking indices, including top journals, countries, and the most frequent keywords in the obtained documents. Out of all articles found, 100 articles that fit the criteria were selected. The main conclusions of the reviewed articles were analyzed and compared accordingly. To focus the literature within the scope of the study, all reviews were categorized into three sections, including general principles for fluoride adsorption, factors influencing the adsorption of this compound, and fluoride adsorption capacity and engineering implications.

3. Adsorption Process for Defluoridation: General Principles

Adsorption is a prominent technique among the various methodologies available for water defluoridation, being one of the most efficient in terms of cost, design, and operation. It is environmentally friendly and offers reliable alternatives with satisfactory results, and is therefore widely used [1–3,16,19,36–42]. Moreover, its additional benefits are the absence of secondary byproducts and hazardous secondary pollutants [43]. Li et al. [44] state that adsorption is the most widely used method for fluoride removal from drinking water, and Ahamad et al. [3] further note that adsorption is applicable for fluoride removal even at low concentrations.

Adsorbent surfaces are often physically and/or chemically heterogeneous and binding energies can vary greatly from one site to another. Thus, one seeks to promote physical adsorption or physisorption, which involves van der Waals forces, and is best suited for a process where the capacity of the adsorbent must be regenerated. Organic and inorganic constituents present in the aqueous phase are efficiently removed by adsorption technology [45]. The efficiency of the process is highly dependent on the physicochemical properties of the adsorbent materials [18]. Thus, although adsorption is a practical and low-cost method for fluoride removal, it needs a mechanically resistant, selective and high-capacity fluoride removal material [14].

Among the factors that influence adsorption, Suriyaraj and Selvakumar [2] cite the surface area of the adsorbent, temperature, pH, nature of the adsorbent and solute, and the stirring speed. In studies based on the adsorption process to F^- removal, it is desirable to have knowledge of these variables and their influence on the adsorption capacity in order to maximize the contaminant removal efficiency [4,8,46–48]. In addition to these factors, the effect of interfering ions should also be considered in F^- removal studies. Naturally, water contains anions such as chloride (Cl^-), nitrate (NO_3^-), sulfate (SO_4^{2-}), bicarbonate (HCO_3^-), carbonate (CO_3^{2-}), and phosphate (PO_4^{3-}), in addition to metal cations, which can compete in the adsorption process and form stable complexes with F^- .

Mourabet et al. [48] studied the response surface methodology (RSM) for the removal of fluoride on Brushite. The authors reported four important process parameters including initial fluoride concentration (40–50 mg L⁻¹), pH (4–11), temperature (10–40 °C) and Brushite dose (0.05–0.15 g). Finding the optimum conditions, a maximum fluoride removal of 88.78% was achieved. Gong et al. [49] studied five alumina's synthesized at different pH and calcination temperatures, and the surface properties and defluoridation performance of alumina were investigated. The authors reported batch experiments and observed that the weak acidic pH favored the fluoride adsorption. Khiewwijit et al. [50] investigated the adsorption of fluoride from an aqueous solution using eggshell pretreated with plasma technology and compared to untreated eggshell. The maximum fluoride removal efficiency increased from 59.88% with untreated eggshell to 90.34%, which was due to an increase in adsorption sites after the surface modification process of the eggshell.

In the mentioned publications we can perceive that the factors that influence the choice of operation mode of the adsorption system, whether continuous or batch, are the volume

to be treated, pH, concentration of adsorbent solid, temperature, and the co-existing ions, as shown in the Figure 1.

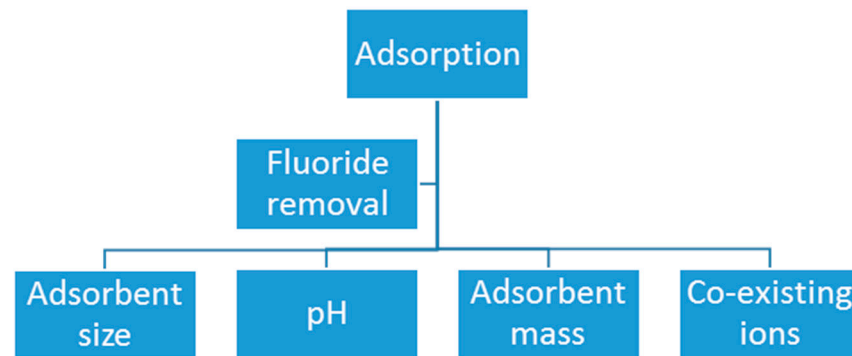


Figure 1. Main parameters to take into consideration in adsorption process.

Some studies related to the adsorption process for defluoridation found in the literature involve two types of systems: (a) bench-scale batch systems, in which a fixed volume of synthetic solution of F^- or natural groundwater is treated, and (b) continuous systems, in fixed or expanded bed columns, in which the volume treated varies with time. When the goal is to treat large volumes, the column system is more suitable, because batching large volumes tends to have a higher implementation cost. Thus, an essential step in adsorption studies is to define the limits of the variables that will be studied. Even for the step of determining the best conditions of the process, it is necessary to define minimum and maximum values of the mass of adsorbent to be applied in a given volume, the initial concentrations of fluoride, the pH range to be studied, and the contact time.

4. Influential Factors in Fluoride Adsorption

4.1. Adsorbent Specific Surface Area and Particle Size

The specific surface area represents the area per unit mass of the adsorbent; thus, a large specific surface area favors adsorption. The adsorption capacity of the adsorbent is expected to be related to the surface area and the number of available active sites. However, the adsorption performance of F^- can also be influenced by different morphologies of the adsorbent and size and shape of microstructures [51]. Table 1 provides a comprehensive survey of studies in the literature focusing on specific surface area (SSA), pore volume (PV), and pore diameter (PD) of adsorbents investigated for the removal of F^- .

Table 1. Specific surface area, pore volume, and pore diameter of adsorbents studied for the removal of F^- .

Reference	Adsorbent Materials	SSA ($m^2 g^{-1}$)	PV ($cm^3 g^{-1}$)	PD (nm)
Azari et al. [17]	Iron oxide	94.96	0.2520	0.584
	Silver oxide	74.34	0.2240	0.421
	Iron and silver oxide (3:1)	254.24	0.6780	0.131
	Iron and silver oxide (1:1)	189.57	0.3910	0.691
	Iron and silver oxide (1:3)	145.74	0.4790	0.621
Babaeiveli and Khodadoust [52]	Titanium oxide powder	56.00	-	20
	Untreated banana peel powder	312.3	-	-
Bhaumik and Mondal [4]	Banana peel powder heat treated	884.7	-	-
	Banana peel powder coated with calcium	912.86	-	-
Cai et al. [51]	Tea residue as support for hydrated aluminum oxide and polyacrylamide	25.86	0.1370	19.76
Cai et al. [53]	Tea residue as support for hydrated aluminum oxide	1.68	0.0083	27.87
Chen et al. [18]	Microspheres and hydroxyapatite	94.1	-	13.8
	Microspheres and hydroxyapatite doped with sulfate	84.2	-	12.8–23

Table 1. Cont.

Reference	Adsorbent Materials	SSA (m ² g ^{−1})	PV (cm ³ g ^{−1})	PD (nm)
Craig, Stillings, and Decker [1]	Activated alumina under varying conditions of surface acidity, hydration period and particle size range	292.1	0.135	-
		288.9	0.153	58
		294.8	0.135	57
		288.4	0.153	-
		285.2	0.136	-
Craig et al. [54]	Activated alumina Bauxite Native laterite	290.2	0.158	59
		288.9	0.35	-
		5.5	0.01	-
Dehghani et al. [55]	Multi-walled carbon nanotubes Single walled carbon nanotubes	21.9	0.02	-
		270	-	-
He et al. [19]	Hydroxyapatite nanowires	700	-	-
Hernandez-Campos et al. [14]	Lanthanide-doped silica xerogels X-B Lanthanide-doped silica xerogels X-LaCl Lanthanide-doped silica xerogels X-La-Cl-M Lanthanide-doped silica xerogels XLaO	77	-	5
		506.94	0.25	1.966
		579.67	0.54	3.736
		509.01	0.55	4.297
Jia et al. [41]	Composite adsorbent based on bayerite and ferrhydrite	732.81	0.81	4.438
Jin et al. [13]	Porous MgO nanoplates	115.95	0.448	-
Kang, Yu, and Ge [56]	Cerium oxide nanorods Cerium oxide octahedra Cerium oxide nanotubes	47.4	-	14.4
		11.10	0.241	3.056
		160.20	0.387	3.054
Karmakar et al. [5]	Metal–organic aluminum fumarate structure	55.80	0.211	3.411
Kumari, Behera, and Meikap [20]	Pure alumina Activated alumina by H ₂ SO ₄	1156	-	-
		74.72	0.214	11.51
Liang et al. [57]	Chitosan magnetic sphere modified with La ³⁺ Chitosan magnetic sphere modified with earth rare	87.44	0.251	11.49
		116.64	-	8.15
Lin, Liu, and Chen [58]	Composite UiO-66-NH ₂	21.24	-	21.24
Ma et al. [59]	Biocarbon lamellar double hydroxide	905	0.43	-
		73.9	-	-
		80.5	-	-
Millar et al. [10]	Activated alumina	87.7	-	-
Mohan, Kumar, and Srivastava [60]	Biocarbon from corn stover with magnetic removal	350	-	-
Mondal, Bhaumik, and Datta [4]	Coconut fiber ash Coconut fiber ash modified with aluminum hydroxide	3.61	-	-
		20.40	-	-
Mukherjee et al. [61]	Nostoc sp. Algae biomass treated with Ca ²⁺ Untreated Nostoc sp. Algae biomass	26.3	-	-
		2.357	0.873	-
Nur et al. [62]	Hydrated ferric oxide	0.993	0.532	-
Raghav and Kumar [12]	Pectin compound loaded with Fe-Al-Ni Alginate compound loaded with Fe-Al-Ni	148	0.214	5.4
		274.59	1.15	109.4
Tomar, Prasad, and Kumar [25]	Composite based on manganese and zirconium	95.73	1.02	58.02
Wan et al. [63]	γ-AlOOH@ CS magnetic nanoparticles	234.973	-	1.12
Wu et al. [64]	Aluminum hydroxide intercalated with oxalate	111.78	-	30–50
Xu et al. [65]	Mesopores alumina (MA-0) Mesopores alumina (MA-glucose—1 mmol) Mesopores alumina (MA-glucose—2.5 mmol) Mesopores alumina (MA-glucose—4.0 mmol) Mesopores alumina (MA-glucose—5.0 mmol)	68	-	7.0
		191	0.54	9.7
		437	0.60	5.0
		321	0.43	5.0
		359	0.44	4.5
		357	0.45	5.0

Pore size is an important factor in the fluoride adsorption mechanism due its effects on the accessibility of the adsorbent material to fluoride ions. In general, the optimal pore size for fluoride adsorption also depends on the size of the fluoride ions [66]. As characteristics, the pore size distribution of the adsorbent material can influence the adsorption capacity,

kinetics, and selectivity of fluoride removal. Adsorbent materials with small pore sizes and narrow pore size distributions can have a high surface area-to-volume ratio, providing many active adsorption sites. However, they can also limit the accessibility of larger fluoride ions to the internal surface of the adsorbent material, which can decrease the adsorption capacity for fluoride [10,20,36,67]. On the other hand, adsorbent materials with larger pore sizes can have a higher accessibility to larger fluoride ions, resulting in a higher fluoride adsorption capacity. However, this may come at the expense of a lower surface area-to-volume ratio and a lower number of active adsorption sites.

Xu et al. [65] studied the gamma phase of mesoporous alumina (MA) synthesized by a hydrothermal method followed by thermal treatment. The as-synthesized MA nanoparticles with an average size of 20 nm–150 nm have an ordered wormhole-like mesoporous structure. The pore size is 5 nm with a narrow distribution, and the specific surface area reaches $357 \text{ m}^2 \text{ g}^{-1}$ while the bulk density is $0.45 \text{ cm}^3 \text{ g}^{-1}$. Glucose as a small-molecule template plays an important role on the morphology, surface area, and pore diameter of the MA.

Wang et al. [68] reported the synthesis of a Fe-La composite adsorbent by one-pot hydrothermal method for removing fluoride and phosphate simultaneously. The as-obtained adsorbent showed rod-like morphology and a large specific surface area of $113.13 \text{ m}^2 \text{ g}^{-1}$. A series of adsorption experiments were carried out to investigate the influence of various factors on the simultaneous adsorption of fluoride and phosphate, such as different adsorbent dosage, pH, temperature, and co-existing anion. The results demonstrate that the Fe-La adsorbent exhibited high adsorption capacity for fluoride and phosphate. Liu et al. [69] studied a magnetic adsorbent consisting of iron–aluminum oxide nanoparticles anchored on graphene oxide (IAO/GO) for fluoride removal from an aqueous solution. By combining, the advantages of graphene oxide (GO) and IAO, IAO/GO exhibits high adsorption capacity (64.72 mg g^{-1}), good acid–alkali stability, super paramagnetism, and good selectivity for fluoride.

Wang et al. [70] investigated the flaky Mg–Fe–La trimetal composite as an adsorbent for fluoride removal synthesized by a facile co-precipitation method. The experimental results demonstrated that the adsorbent, with an Mg/Fe/La molar ratio of 25:1:4, obtained the largest adsorption capacity of 112.17 mg g^{-1} for fluoride. The BET surface area and average pore diameter of the adsorbent were $178.55 \text{ m}^2 \text{ g}^{-1}$ and 31.16 nm, respectively.

Upon reviewing the cited studies on fluoride adsorption (Table 2), a notable discrepancy in specific surface area values was observed, attributed to variations in the types of adsorbents employed for defluoridation. However, establishing a direct correlation between specific surface area values and adsorption capacity or fluoride removal efficiency proved challenging within the scope of this literature review. It is noteworthy that, despite the disparity in specific surface area values, it does not emerge as the primary factor influencing adsorption capacity.

Craig et al. [1] studied the physicochemical properties and fluoride adsorption capacity of activated alumina (AA) under varying conditions of surface acidity, hydration period, and particle size range. The authors showed that the overall adsorption does not change notably between the two particle size ranges analyzed, but adsorption is much faster when the particles are reduced from 0.5–1.0 mm to 0.125–0.250 mm. At the larger particle size, AA adsorbs about 10% of fluoride from the solution at 5 min, requiring 24 h to remove 90%. In contrast, the thinner AA adsorbs almost 60% fluoride in 5 min and 90% after 40 min. The authors attributed this result to the inverse relationship between diffusion rate and particle size. They further argue that the particle size used in a defluorination filter should be as fine as reasonably possible while maintaining an acceptable flow rate, i.e., to increase the adsorption rate, a relatively fine particle size is recommended.

In conclusion, while specific surface area variations among adsorbents were observed, their direct impact on adsorption capacity remains inconclusive. However, the influence of particle size on fluoride adsorption capacity is evident, with practical recommendations provided for optimizing the adsorption process. Further research is warranted to explore

the intricate interplay between specific surface area, particle size, and overall adsorption performance in defluoridation applications.

Table 2. Adsorbents and temperature ranges studied in exothermic processes of F^- removal by adsorption.

Reference	Adsorbent Materials	Temperature
Ali, Alothman, and Sinagi [28]	Iron nanoparticles impregnated via green technology	20–30 °C
Azari et al. [17]	Iron and silver oxide nanoadsorbent	20–50 °C
Bahumik and Mondal [4]	Banana peel powder	40–70 °C
Gao et al. [71]	Mg, Al, and lamellar double hydroxide immobilized on magnetic alginate nanoflakes	-
Jia et al. [41]	Composite adsorbent based on ferrhydrite and bayerite	25–35 °C
Karmakar et al. [5]	Metal–organic structure of aluminum fumarate	20–60 °C
Lin, Liu, and Chen [72]	Composite UiO-66-NH ₂	20–60 °C
Mohan, Kumar, and Srivastava [60]	Corn stover biocarbon with and without magnetic removal	25–45 °C
Modal, Bhaumik, and Datta [4]	Coconut fiber ash impregnated with aluminium hydroxide	40–100 °C
Tang and Zhang [68]	Fe (III) bimetallic oxide composite	25–35 °C
Wang et al. [73]	Zirconium imobilized in sodium carboximetilcelulose	25–55 °C

4.2. The Influence of Temperature on the Adsorption of F^-

The influence of temperature on adsorption processes is related to the kinetic energy, mobility, solubility, and chemical potential of the adsorbate. Thus, a change in the temperature of a process will lead to a change in the adsorption capacity. Generally, an increase in temperature can enhance the fluoride adsorption capacity, but it can also decrease the adsorption rate.

The increase in fluoride adsorption capacity with increasing temperature may be due to several factors. For example, higher temperatures can increase the mobility of the fluoride ions and their diffusion towards the adsorbent material, increasing the chances of successful adsorption. Additionally, higher temperatures can increase the surface area of the adsorbent material due to thermal expansion, resulting in a larger number of active adsorption sites [23,58]. Table 2 presents the adsorbents and temperature ranges studied in exothermic processes of F^- removal by adsorption.

The temperature rise probably causes a reduction in the thickness of the boundary layer [4], and the solubility of the F^- [4]; thus, the F^- ions tend to escape from the adsorbent surface (solid phase) to the solution (liquid phase), which results in a reduction in adsorption capacity as the temperature rise is increased. Another possibility pointed out by Mondal et al. [8] is that an increase in the thermal energy of the adsorbed F^- occurs at higher temperatures, which ultimately causes an increase in desorption.

On the other hand, the decrease in the adsorption rate with increasing temperature may be due to a decrease in the adsorption affinity between the adsorbent material and the fluoride ions. This can occur when the adsorption process is exothermic, meaning that heat is released during the adsorption process. The release of heat can cause a decrease in the driving force for adsorption, resulting in a decrease in the adsorption rate [23,74]. Some studies were found in which it is reported that the adsorption of F^- increases with increasing temperature, which is an indication that the adsorption process is endothermic. Table 3 presents the adsorbents and the temperature ranges studied by these authors.

Table 3. Adsorbents and temperature ranges studied in endothermic processes of F[−] removal by adsorption, considering the isosteric enthalpy.

Reference	Adsorbent Materials	Temperature
Chen et al. [57]	Hydroxyapatites microspheres doped with sulfate	25–45 °C
Rojas-Mayorga et al. [75]	Activated carbon from bone, synthesized via metal doping, using Al and Fe salts	25–40 °C
Wan et al. [63]	γ -AlOOH@Cs magnetic nanoparticles	20–50 °C
Nahum et al. [76]	Bone char made from cattle bones	15–35 °C
Valdivieso et al. [77]	α -Al ₂ O ₃	10–40 °C

According to Ghosh et al. [78], the temperature elevation probably causes an increase in thermally accessible adsorption sites for the adsorption of F[−], by pore unclogging, on the surface of the adsorbents. It is concluded that, of the 17 papers analyzed showed in the tables, it was found in 12 of them that an elevation of temperature up to 60 °C during the process does not bring significant improvement for the adsorption capacity of F[−]. It is also emphasized that this finding is very important for the use of adsorption for defluoridation, since a real system should operate at room temperature, and in this system the increase in temperature for an eventual gain in adsorption capacity is not economically feasible.

Valdivieso et al. [77] studied the effect of temperature and pH on the zeta potential of α -Al₂O₃ and adsorption of fluoride ions. The authors reported that when the temperature increases from 10 to 40 °C, the pH of the point of zero charge (pH_{PZC}) shifts to smaller values, indicating proton desorption from the alumina surface. The pH_{PZC} increases linearly, allowing the estimation of the standard enthalpy change for the surface-deprotonation process. In addition, the authors reported that the increasing of the temperature from 25 to 40 °C lowers the adsorption density of fluoride.

Nahum et al. [76] studied the effects of temperature on the adsorption of fluoride onto bone char made from cattle bones. The authors argue that the adsorption capacity was not influenced by temperature in the range from 15 to 35 °C. A comparison of fluoride adsorption capacities among several adsorbents revealed that the adsorption capacity of the bone char was 2.8 and 36 times greater than those of a commercial activated alumina (F-1) and a commercial activated carbon (F-400).

Adsorption is commonly associated with exothermic processes, where heat is released during adsorption [79]. This phenomenon is consistently observed across various adsorption systems [80]. On the other hand, the occurrence of endothermic adsorption is less common and may raise questions about the thermodynamic feasibility of the process.

In endothermic processes, the entropy of the system must increase for the reaction to occur spontaneously, according to the second law of thermodynamics [81]. This may seem contradictory, as the transition of the adsorbate from the solvated state to the adsorbed state is usually accompanied by a decrease in entropy. Therefore, the occurrence of endothermic adsorption may appear implausible at first glance. However, it is essential to consider that other factors, such as changes in the system's structure and free energy, can influence the feasibility of adsorption [26].

In some cases, despite the decrease in entropy during adsorption, other thermodynamic contributions, such as variations in enthalpy, can offset this effect and allow the process to occur endothermically. Additionally, specific experimental conditions, such as the presence of solvents or other components in the system, can significantly influence the thermodynamic behavior of adsorption [82].

4.3. The Influence of pH on the Adsorption of F[−]

The determination of a favorable pH value is a fundamental step, as this controls the entire chemical defluorination process [11,62]. The pH interferes at the solid/liquid interface by changing the surface charges and functional groups, modifying the surface properties and active sites of the adsorbents [16,17]. Moreover, most adsorbents used for F[−] removal are applicable only in narrow pH ranges [83].

The first step in identifying the favorable pH range is the determination of the zero-charge point (pH_{PZC}). The balance of electrical charges on the surface of the adsorbent can be positive, negative, or zero. The pH_{PZC} is defined as the pH at which the net charge on the adsorbent surface is zero. If $\text{pH} < \text{pH}_{\text{PZC}}$, the surface charge is positively charged by the excessive amount of H^+ ions, and consequently adsorption of anions is favored. If $\text{pH} > \text{pH}_{\text{PZC}}$, the surface charge is negatively charged by OH^- ions, and consequently adsorption of cations is favored [15,20].

Table 4 presents pH_{PZC} values from F^- removal studies by adsorption, and the pH at which the best fluoride removal conditions were obtained, i.e., the highest removal efficiency or adsorption capacity values.

Table 4. Values of pH_{PZC} and the pH of highest efficiency cited in F^- removal studies.

Reference	Adsorbent Material	pH_{PZC}	pH_{ideal}
Ali, Alothman, and Sanagi [28]	Iron nanoparticles impregnated via green technology	-	7
Azari et al. [17]	Iron oxide	5.9	-
	Silver oxide	7.1	-
	Iron and silver oxide (3:1)	6.0	3
	Iron and silver oxide (1:1)	6.7	-
	Iron and silver oxide (1:3)	6.1	-
Babaeivelni and Khodadoust [52]	Crystalline titanium dioxide powder	6–6.5	2–5
Barathi, Kumar, and Rajesh [23]	Zirconium impregnated in cellulose matrix	-	4.5–5.5
Bahumik and Mondal [4]	Untreated banana peel powder	6.2	5.6
	Banana peel powder treated heat termic	8.1	6.1
	Banana peel powder coated with calcium	8.2	7.2
Cai et al. [53]	Tea residue supported to hydroxide aluminum oxide and anionic polyacrylamide	-	4–9
	Tea residue as a support medium for Al and Fe oxides	-	4–8
Cheng et al. [18]	Activated alumina	8.9	5.84
	Activated alumina modified with lanthanum nitrate	9.6	4.99
Craig et al. [1]	Activated alumina	8	5.5–6.5
	Activated alumina modified with CaO	11.9	4–10
Dayananda et al. [84]	Mesoporous alumina modified with CaO	8.2	-
	Mesoporous alumina	-	5
Dong and Wang [40]	Magnetic cationic hydrogel coated with lanthanum	7	2.8–5.2
Gao et al.	Mg, Al, and lamellar double hydroxide immobilized in alginate magnetic nanoflakes	9	5
Gong et al. [49]	Activated alumina (A1) PZC 1	10.1	5.5–6.7
	Activated alumina (A2) PZC 1	9.9	
	Activated alumina (A3) PZC 1	9.7	
	Activated alumina (A4) PZC 1	9.4	
	Activated alumina (A5) PZC 1	9.2	
	Activated alumina (A1) PZC 2	9.7	
	Activated alumina (A2) PZC 2	9.7	
	Activated alumina (A3) PZC 2	9.5	
	Activated alumina (A4) PZC 2	9.5	
	Activated alumina (A5) PZC 2	9.4	
	Activated alumina (A1) PZC 3	4.8	
	Activated alumina (A2) PZC 3	5.3	
	Activated alumina (A3) PZC 3	6.5	
	Activated alumina (A4) PZC 3	6.8	
	Activated alumina (A5) PZC 3	7.6	

Table 4. Cont.

Reference	Adsorbent Material	pH _{PZC}	pH _{ideal}
Kang et al. [56]	3 different cerium oxide morphologies (nanocubes, octahedra, nanorods)	-	3
Karmakat et al. [5]	Aluminum fumarato organic–metal structure	8.1	2–7
Kumari, Behera, and Meikap [20]	Alumina activated to H ₂ SO ₄	8.2	6–7
Mohan, Kumar, and Srivastava [60]	Biocarbon from corn stover with magnetic removal Biocarbon from corn stove	1.96 2.01	2
Mondal, Bhaumik, and Datta [8]	Coconut fiber ash impregnated with aluminum hydroxide	7.2	5
Mukherjee et al. [61]	Untreated Nostoc Sp. algae biomass	7.2	5
Mullick and Neogi [15]	Powder activated carbon and zirconium	5.44	2
	Powder activated carbon and magnesium	9.44	
	Powder activated carbon and manganese	6.23	
	Mg, Zr, and Mn impregnated in powder Activated carbon via ultrasound assisted sintese	11.90	
Nur et al. [62]	Hydrated ferric oxide	4–5	3
Vences-Alvarez et al. [85]	Comercial granular activated carbon	8.7	5–6
	Comercial granular activated carbon modified with lanthanum	8.9	

Ansari et al. [17] showed that the removal of F[−] using multi-walled carbon nanotubes was highly dependent on the pH of the solutions. The maximum adsorption of 93.5% was obtained at pH 5. For higher pH values, the adsorption capacity decreased significantly, reaching 41.2% at pH 9. This sharp reduction in the removal of adsorbed F[−] in the alkaline pH range was attributed to competition for adsorption sites between hydroxyl ions and F[−].

Babaeivelni and Khodadoust [52] studied the effect of pH (in the range of 2 to 11) on the adsorption of F[−] on crystalline titanium dioxide (TiO₂) powder. The maximum uptake of F[−] was observed at acidic pH (pH 2), with an adsorption capacity of approximately 0.2 mg g^{−1}. The adsorption capacity was reduced to 0.16 mg g^{−1} at pH 7, to 0.14 mg g^{−1} at pH 9, and was only 0.02 mg g^{−1} at pH 11. The reduction in F[−] uptake in the alkaline solution was attributed to competition between hydroxyl groups and F[−] ions for the active sites on the adsorbent. On the other hand, the positively charged surface sites in the acidic solution increased the F[−] uptake. The final F[−] concentration in the pH 7–8 range was almost at the WHO recommended limit of 1.5 mg L^{−1}, while the F[−] concentration was less than 1 mg L^{−1} in the acidic solutions.

Barathi et al. [23] used zirconium hydroxide impregnated on a cellulose matrix for F[−] removal. In the pH range of 4.5 to 5.5, the adsorption of F[−] on the surface of the adsorbent is quite efficient. The authors argued that in a weakly acidic medium, the F[−] ion can interact with the positively charged surface hydroxyl groups of cellulose. Above pH 5.5, deprotonation of the hydroxyl groups on the surface of the adsorbent the competition between OH[−] and F[−] for the active adsorption sites reduced the efficiency of the process.

Cai et al. [39] developed a hybrid adsorbent by impregnating Li/Al LDHs (lamellar double hydroxides) on the commercial anion exchanger D201 (LALDH-201), and analyzed the effect of pH. LALDH-201 exhibited a higher adsorption capacity than D201 and activated alumina (AA). The adsorption capacity of LALDH-201 increased rapidly from pH 2 to 4 and decreased more slightly from pH 4 to pH 9. It then decreased sharply from pH 9 to 12. According to the authors, this pH-dependent adsorption can be interpreted by their different adsorption mechanisms, which were (1) ion exchange between the F[−] and covalent groups of the D201 matrix, (2) ion exchange between the F[−] and exchangeable anions of the Li/Al LDHs, and (3) ligand exchange to form an internal sphere complex, resulting in the release of the hydroxyl ion. The adsorption capacity of both LALDH-201 and D201 decreased at alkaline pH (> 9.0), possibly due to competition between OH[−] and

F^- for the exchangeable sites. In addition, the higher concentration of hydroxyl ions in alkaline solution further restricts the F^- adsorption capacity of LALDH-201. The reduction in adsorption observed for D201 below pH 4 is because F^- forms weakly ionized HF in the acidic solution. For LALDH-201, the decrease in adsorption capacity below pH 4 was attributed as a result of more than one factor: with the formation of weakly ionized HF, LALDH-201 becomes unstable at pH 3.5, which is also unfavorable for the adsorption of F^- . The capacity of D201 was relatively stable at pH 4 to 9, where the concentration of H^+ or OH^- was insufficient to affect ion exchange. For AA, the results indicated that its F^- adsorption capacity was reduced at low and high pH, with the optimum pH at 5–6.

Cai et al. [86] synthesized hybrid adsorbents of Li/Al-LDHs doped with lanthanum (La). The highest adsorption capacity values were observed at pH = 7, being approximately 31.5, 17.5, and 3.5 mg g⁻¹ for LAL03, LA3, and AA, respectively. LAL03 is positively charged at pH = 7, and the authors attribute this to probable formation of Al-OH²⁺ and La-OH²⁺, which is favorable for removal of anionic pollutants such as F^- . pH values < 4 were not considered due to the formation of weakly ionized HF. When the pH was increased to 8, a reduction in the adsorption of F^- was observed, which was attributed to competition between OH^- and F^- for the exchange sites. For pH > 10, dissolution of aluminum from the layered structure is attributed as a likely cause of the reduced adsorption capacity. The authors further argue that AA can be negatively affected by hydroxyl ions at high pHs, which resulted in its rather narrow optimal pH range of 5 to 6.

Chai et al. [83] demonstrated that a sulfate-doped Fe₃O₄/Al₂O₃-based composite adsorbent is positively charged for pH values < pHPZC, which benefits the adsorption of F^- . Thus, the adsorbent exhibited a high adsorption efficiency (from 90 to 70%) over a wide pH range (4 to 10). For pH > pHPZC, the surface of the nanoparticles was negatively charged, which tends to repel the F^- ions, reducing adsorption. Consequently, at pH > 11.2 the removal efficiency decreased considerably. The low adsorption efficiency observed at pH < 4 was attributed to the formation of HF, which reduced the electrostatic attraction between the F^- and the adsorbent surface.

Kumari et al. [20] studied the effect of pH on the removal of F^- by alumina by acid activation in H₂SO₄ (AAA). The authors observed that the extreme ranges of lower and higher pH were unfavorable for the adsorption of F^- by AAA. Very low pH promotes the formation of weak hydrofluoric acid as well as AlF²⁺ and AlF⁺ complexes that lead to decreased F^- adsorption. In the basic region the reason for reduced F^- adsorption is the competitive environment between F^- ions and OH^- to adsorb onto AAA. Both adverse effects were negligible in the pH range of 6–7, which contributed to the maximum adsorption of F^- on AAA, which was approximately 97%. Thus, pH 6.5 was the optimal pH.

The analysis of 56 articles reveals a significant pH dependency in the adsorption process of F^- , with a stable adsorption capacity in acidic conditions favoring protonation. This results in a positively charged surface on adsorbents, enhancing attraction forces with F^- ions. Notably, adsorbents for F^- removal tend to be effective within a narrow pH range, with only 17 papers highlighting efficiency in alkaline conditions. Given that groundwater typically has a slightly alkaline pH, selecting an adsorbent efficient in this range is crucial for effective defluoridation. Additionally, both lower (pH < 3–4) and higher (pH > 9–10) pH ranges are unfavorable for F^- adsorption. At lower pH, weakly ionized HF formation reduces adsorption, while at higher pH, negative solution charge leads to repulsion between negatively charged adsorbents and F^- ions. Competition with OH^- ions further diminishes adsorption. Hence, water pH must be considered when choosing an adsorbent for fluoride removal, and optimizing pH during treatment is vital for maximum efficiency.

Hence, careful consideration of water pH is essential when choosing an adsorbent for fluoride removal. Optimizing the water's pH during treatment is crucial to ensure peak efficiency. Additionally, the awareness of pH variations over time and across different locations underscores the necessity for ongoing monitoring throughout the treatment

process. This consistent oversight is vital to guarantee a reliable and effective fluoride removal process.

4.4. The Effect of Adsorbent Mass on the Removal of F^-

The optimal concentration of the adsorbent material for fluoride removal depends on several factors, including the specific adsorbent material used, the initial concentration of fluoride in the water, and the desired level of fluoride removal. Table 5 presents an overview of F^- removal studies by adsorption considering the influence of adsorbent mass.

Table 5. Overview of F^- removal studies by adsorption considering the adsorbent mass.

Reference	Adsorbent Materials	Mass (g L ⁻¹)	qe (mg g ⁻¹)	Removal (%)
Ali, Allothman, and Sanagi [28]	Iron nanoparticles impregnated via green technology	2.5	1.44	90
Azari et al. [17]	Silver and iron oxide (3:1)	0.5	20	100
Babaeivelni and Khodadoust [52]	Crystalline titanium dioxide powder	25	153.5	75
Bhaumik and Mondal [4]	Zr impregnated in cellulose matrix	10–12	-	-
	Untreatment banana peel powder	1	17.4	69.4
	Banana peel powder heat treatment	1	26.3	81.3
	Banana peel powder coated with calcium	1	39.5	82.6
Cai et al. [51]	Tea residue as a support for hydrated aluminum oxide	1	-	71.5
	Tea residue as a support for hydrated aluminum oxide and anionic polyacrylamide	1.2	7.43	89.7
Cai et al. [53]	Tea residue	8	-	30
	Tea residue as a support for iron oxide	8	-	60
	Tea residue as a support for aluminum oxide	0.4–2	-	90
	Tea residue as a support for silver and iron oxide	0.4–2	-	90
Dayananda et al. [84]	Mesoporous alumina (Al ₂ O ₃) modified with CaO	3	1.48	90
	Mesoporous alumina	3.0	1.38	56
Mullik and Neogi [15]	Mg, Mn, and Zn impregnated in powder carbon activated via ultrasound assisted sintese	1	-	96
Rafique et al. [87]	Immobilized modified activated alumina	10	0.48	95
Raghav and Kumar [88]	Pectin compound coated with Fe-Al-Ni	0.5	10.91	98.2
	Alginate compound coated with Fe-Al-Ni	0.5	2.3	92
Tang and Zhang [89]	Fe and Ce (IV) bimetallic oxide compound	0.5	-	94.73
Thakur et al. [29]	Magnetic compound with Fe ₃ O ₄ nanoparticles encapsulated in Zr (IV) net and polyacrylamide	6	-	95
Wang et al. [73]	Zr (IV) immobilized in sodium carboximetilcelulose	8.0	-	90

According to Table 5, it is evident that the concentration of the adsorbent material directly influences the efficiency of fluoride removal in the adsorption process. Elevated concentrations of the adsorbent material can enhance the number of active adsorption sites and expand the available surface area for fluoride adsorption. Nevertheless, excessively high concentrations may result in overcrowding of adsorption sites, leading to a reduction in efficiency due to mass transfer limitations. Additionally, the economic aspect plays a crucial role, as the cost of the adsorbent material may pose limitations on determining the optimal concentration.

The balance lies in optimizing the concentration of the adsorbent material for fluoride removal, considering the specific conditions of the treatment process. Striking this balance is crucial not only for efficient removal but also for ensuring economic feasibility, especially in large-scale applications where higher concentrations may incur substantial costs. Therefore,

a thoughtful consideration of the optimal concentration is essential for achieving effective fluoride removal while managing economic constraints.

Ali et al. [17] applied masses of an iron nanoparticle-based composite adsorbent ranging from 0.5–5.0 g L⁻¹. The authors obtained the maximum adsorption capacity of 1.44 mg g⁻¹ for the mass of 2.5 g L⁻¹, with which a percentage removal of F⁻ of 90% was obtained. Babaeiveli and Khodadoust [52] studied the effect of masses from 10 to 50 g L⁻¹ of the adsorbent crystalline titanium dioxide powder (TiO₂) on the adsorption of F⁻. The adsorption increased with increasing mass up to 25 g L⁻¹. For masses from 37.5 to 50 g L⁻¹, no increase in adsorption was observed because saturation was obtained. Barathi et al. [23] used masses of zirconium hydroxide impregnated in a cellulose matrix ranging from 6 to 12 g L⁻¹, with the removal of F⁻ being effective in the range of 10 to 12 g L⁻¹. The authors attribute the increased adsorption to the strong electrostatic attraction between F⁻ and the biopolymer adsorbent. Beyond 12 g L⁻¹, the active adsorption sites were saturated and there was no significant change in the percentage of adsorption.

A further increase in mass (>2.5 g L⁻¹) did not result in any increase in adsorption capacity. Azari et al. [17] demonstrated that with increasing the mass of iron oxide and silver (3:1) from 0.1 to 1 g L⁻¹, the percent removal increased from 26.21 to 100%, and the adsorption capacity decreased from 26.21 to 10 mg g⁻¹. According to the authors, this happened because an increase in the amount of adsorbent provides a greater number of active sites available for binding of F⁻. The highest adsorption efficiency of F⁻ was obtained with the mass of 1 g L⁻¹; however, considering the adsorbent cost, the authors adopted the use of 0.5 g L⁻¹ as it was a more economical value. Bhaumik and Mondal [4] used banana peel powder for the adsorption of F⁻, varying the mass from 0.01 to 1.0 g L⁻¹. The adsorption percentage increased with increasing adsorbent mass, and the highest adsorption percentages were 69.4, 81.3, and 82.6% for BPD-1 (untreated), BPD-2 (heat-treated), and BPD-3 (calcium-coated), respectively. It was observed that, up to a certain level, higher masses resulted in higher F⁻ removals. This was attributed to the higher availability of surface area and pore volume.

Gong et al. [49] studied the mass achievement of five different types of alumina in defluoridation. To reduce the F⁻ concentration of natural water below 1.0 mg L⁻¹, the required masses of A1, A2, A3, A4, and A5 were 0.2, 0.3, 0.5, 0.5, and 1 g L⁻¹, respectively. In contrast, for the commercial Al(OH)₃ and Al₂O₃ adsorbents, the mass was 5 g L⁻¹ for both, with the F⁻ being reduced to 1.86 and 1.28 mg L⁻¹, respectively.

Hu et al. [90] studied the effect of the mass of Fe₃O₄/chitosan/Al(OH)₃ magnetic beads in the range of 0.05–0.25 g L⁻¹. For the masses of 0.05 and 0.10, the adsorption capacities were 23.5 and 35.5 mg g⁻¹, respectively. Thus, the adsorption capacity increased with increasing adsorbent mass. According to the authors, this occurred because the adsorption sites increased as the mass increased. For the masses of 0.15, 0.20, and 0.25 g L⁻¹, the adsorption capacities were approximately 31, 23.5, and 18.5 mg g⁻¹, respectively. Thus, the adsorption capacity decreased with increasing mass. Therefore, an excess of adsorbent reduces the adsorption efficiency, with the best mass being 0.10 g L⁻¹.

Kumari et al. [20] found that an increase in alumina mass by acid activation in H₂SO₄ from 2 to 26 g L⁻¹ increases the F⁻ removal efficiency from 50.2 to 99.2%, but reduces the adsorption capacity from 10.15 to 0.32 mg g⁻¹. The optimum mass found was 14 g L⁻¹. The authors point out two probable causes for this variation: (i) the constant concentration of F⁻ that was used (40 mg L⁻¹), and (ii) the constant active sites on the adsorbent per unit mass. With increasing adsorbent mass, the number of sites available for adsorption of F⁻ increases, so more F⁻ is removed. But, with the further increase in adsorbent mass, there is no utilization of the excess adsorbent available per unit mass, which contributes to the reduction in adsorption capacity. Consequently, less utilization efficiency per unit mass of the adsorbent occurs. Tang and Zhang [89] analyzed the effect of mass from 0.3 to 1.5 g L⁻¹ of Fe (III) and Ce (IV) bimetallic oxide composite adsorbent on F⁻ removal. The removal of F⁻ increased from 85.74 to 94.73% with the mass of adsorbent increasing from 0.3 to

0.5 g L⁻¹. No further increase in F⁻ removal rate was observed with further increase in adsorbent mass. Thus, 0.5 g L⁻¹ was set as the optimum mass.

Following the analysis of the presented papers investigating the impact of adsorbent mass on F⁻ removal, a consistent trend emerges. Irrespective of the specific adsorbent employed, an increase in the quantity of adsorbent leads to a greater availability of active sites for F⁻ binding. However, a critical threshold is observed, beyond which further increments in mass yield diminishing returns. Beyond this point, excess adsorbent does not contribute to increased utilization efficiency per unit mass, and the adsorption capacity begins to decline. This decline is attributed to the decreasing concentration of remaining F⁻, resulting in diminished utilization efficiency.

4.5. The Effect of Co-Existing Ions on the Removal of F⁻

Co-existing ions can compete with fluoride ions for adsorption sites on the surface of the adsorbent material, reducing the overall fluoride removal efficiency. Table 6 presents the co-existing ions with the most and least interference in the adsorption of F⁻ found in the literature review.

Table 6. Co-existing ions with greater and lesser interference in the adsorption of F⁻.

Reference	Co-Existing Ions	
	High Interference	Low Interference
Ali, Allothman, and Sanagi [28]	Na ⁺ , K ⁺ , Ca ²⁺ , Mg ²⁺	-
Azari et al. [17]	PO ₄ ³⁻ , HCO ₃ ⁻	SO ₄ ²⁻ , NO ₃ ⁻ , Cl ⁻
Cai, J. et al. [53]	PO ₄ ³⁻ > SO ₄ ²⁻ > HCO ₃ ⁻ > Cl ⁻ > NO ₃ ⁻	-
Chai et al. [83]	CO ₃ ²⁻ , HCO ₃ ⁻ , PO ₄ ³⁻	SO ₄ ²⁻ , NO ₃ ⁻ , Cl ⁻
Chen et al. [60]	HCO ₃ ⁻ , PO ₄ ³⁻	SO ₄ ²⁻ , NO ₃ ⁻ , Cl ⁻
Dayananda et al. [84]	HCO ₃ ⁻	SO ₄ ²⁻ , NO ₃ ⁻ , Cl ⁻
Dong and Wang [40]	HCO ₃ ⁻ , SiO ₄ ²⁻	SO ₄ ²⁻ , NO ₃ ⁻ , Cl ⁻
He et al. [19]	CO ₃ ²⁻ , HCO ₃ ⁻ , PO ₄ ³⁻	SO ₄ ²⁻ , NO ₃ ⁻ , Cl ⁻
Hu et al. [90]	HCO ₃ ⁻ > CO ₃ ²⁻ > SO ₄ ²⁻ > NO ₃ ⁻ > Cl ⁻	-
Jia et al. [41]	HCO ₃ ⁻	NO ₃ ⁻ , SO ₄ ²⁻ , CO ₃ ²⁻ , Cl ⁻
Jin et al. [13]	CO ₃ ²⁻ , HCO ₃ ⁻ , PO ₄ ³⁻	SO ₄ ²⁻ , NO ₃ ⁻ , Cl ⁻
Karmakar et al. [5]	CO ₃ ²⁻ , SO ₄ ²⁻	HCO ₃ ⁻ , Cl ⁻
Liang et al. [57]	HCO ₃ ⁻ , CO ₃ ²⁻	SO ₄ ²⁻ , NO ₃ ⁻ , Cl ⁻
Mondal, Bhaumik, and Datta [8]	SO ₄ ²⁻ , PO ₄ ³⁻	NO ₃ ⁻ , Cl ⁻
Nur et al. [62]	PO ₄ ³⁻ > SO ₄ ²⁻ > NO ₃ ⁻ > Cl ⁻	-
Wu, S. et al. [64]	CO ₃ ²⁻ , HCO ₃ ⁻ , PO ₄ ³⁻	SO ₄ ²⁻ , NO ₃ ⁻ , Cl ⁻

The degree of competition in the adsorption process is contingent upon the concentration and chemical characteristics of co-existing ions. Naturally occurring water includes anions like chloride (Cl⁻), nitrate (NO₃⁻), sulfate (SO₄²⁻), bicarbonate (HCO₃⁻), carbonate (CO₃²⁻), and phosphate (PO₄³⁻), along with metal cations. These co-existing ions can vie for adsorption sites and form stable complexes with F⁻. Table 6 illustrates that certain co-existing ions, such as chloride and sulfate, share similar ionic radii and charges with fluoride ions, establishing them as robust contenders for adsorption sites. Additionally, other co-existing ions, such as bicarbonate and carbonate, can create complexes with the surface of the adsorbent material. This interaction diminishes the number of active adsorption sites available for fluoride adsorption [8,51,91]. Therefore, the impact of co-existing

ions is contingent on their concentration and chemical properties, as well as the nature of the adsorbent material employed.

Azari et al. [17] highlight that the effect of bicarbonate can be explained by the fact that conjugate bases of weak acids produce more hydroxyl ions, which compete with F^- for adsorption sites. Cai et al. [51] point out that the reduction in the adsorption capacity of F^- with increasing bicarbonate concentration is due to the increase in pH in the solution. To Nur et al. [62], the adsorption mechanisms of phosphate cause this ion to be strongly adsorbed on metal oxides and high valence hydroxides being therefore able to compete easily with F^- . However, Chai et al. [83], Ghosh et al. [84], and Wan et al. [63] clarify that considering the typical range of natural phosphate concentration in groundwater (ranging from 0 to 5 mg L⁻¹), is very low, hence the effect of this ion on F^- adsorption should hardly be a limiting factor for groundwater defluoridation. According to He et al. [19] and Hu et al. [90] HCO_3^- , CO_3^{2-} , and PO_4^{3-} significantly affect F^- removal because these ions are hydrolyzed, leading to an increase in pH value. As a result, more OH^- ions can compete with F^- for the active adsorption sites, which results in decreased adsorption capacity.

On the other hand, the NO_3^- , SO_4^{2-} , and Cl^- ions interfered the least with F^- removal. Nur et al. [62] clarifies that sulfate competes with F^- better than Cl^- and NO_3^- . Nitrate and chloride, on the other hand, are unable to compete well with F^- when they are at equal concentrations. According to the authors, this order of anion competition is like that reported in other studies. According to Karmakar et al. [5], anions that are not specifically adsorbent, such as nitrate and chloride, do not interfere with the adsorption process. For Chen et al. [92], SO_4^{2-} does not significantly reduce the adsorption of F^- , due to its larger ionic radius (2.30 Å). Chloride and nitrate do not significantly reduce the adsorption of F^- because these two ions have a lower binding affinity for the active sites. In addition, these anions have a larger ionic radius than F^- .

Kumar et al. [93] investigated the role of $ZnCl_2/FeCl_3$ -rice husk-modified biochar (Zn-BC and Zn/Fe-BC) in treating F^- contaminated surface and groundwater under the influence of varying solution chemistry, co-existing ions, and biochar-amended through column transport experiments. Modified biochar showed maximum F^- adsorption, 99.01% and 91.90% using Zn/Fe-BC and Zn-BC, respectively, compared to 85.87% using raw biochar (R-BC). They also found that the increased salt strengths led to reduced electrophoretic mobility of biochar particles, i.e., biochar–biochar particles attract each other and increase the hydrodynamic diameter, which ultimately reduces the active sites on biochar for F^- adsorption. The authors argue that co-transport and deposition of biochar and F^- in saturated porous media revealed lower mobility of biochar, and maximum F^- adsorption was observed at 10 mM salt strength. Biochar transport is governed by electrostatic interactions, whereas F^- transport mainly occurs through chemisorption.

Wang et al. [94] studied bovine bone meal (BBM) used as a green adsorbent for Cd^{2+} and F^- adsorption in single and binary solutions. The results showed that with the increase in pH, the uptake ability of BBM towards Cd^{2+} and F^- showed an upward and downward trend, respectively, suggesting that electrostatic interaction was an important mechanism. Cl^- and NO_3^- had little effect on Cd^{2+} and F^- adsorption, while $H_2PO_4^-$ enhanced adsorption capacities.

Upon analyzing the cited articles, a clear consensus emerges, highlighting that HCO_3^- , CO_3^{2-} , and PO_4^{3-} ions exert the most pronounced impact on the adsorption of F^- , leading to a reduction in adsorption capacity. This decline is attributed to the competitive nature of adsorption between F^- ions and these interfering ions. Azari et al. [17] further contend that the influence of these anions on the adsorption process may be linked to their affinity with the adsorbents. Additionally, insights from Cai et al. [39] and Raghav et al. [88] emphasize a broader perspective, indicating that, in general, higher valence anions tend to have a more substantial interfering effect on the defluoridation process. This suggests that the interference level is not only a function of the specific interfering ion but is also influenced by the valence state of the ions involved.

In summary, using adsorption for fluoride removal in water faces various challenges, such as limitations in adsorption capacity, pore size, temperature sensitivity, interference from co-existing ions, pH dependency, optimal solid dosage, and cost considerations. It is crucial to carefully assess these constraints and explore alternative treatment methods to determine the most appropriate approach for specific applications. Despite these challenges, recognizing the environmental potential of adsorption for fluoride removal is crucial. Investigating fluoride adsorption parameters is essential for optimizing processes and minimizing the environmental impact of water treatment. Addressing these limitations and advancing our understanding of fluoride adsorption parameters contributes to more sustainable and efficient water treatment practices.

5. Fluoride Adsorption Capacity and Engineering Implications

The adsorption capacity of the adsorbent (q_e) represents the amount of adsorbate adsorbed per unit mass of adsorbent at equilibrium and is the main parameter needed for the design of an adsorption system. This parameter plays a central role in the design and optimization of an adsorption system. In the context of fluoride removal, Table 7 provides a comprehensive overview of recent works, presenting values of adsorption capacity alongside the corresponding conditions under which fluoride removal occurred.

Table 7. Adsorption capacity obtained in studies of F^- removal by adsorption.

Reference	Adsorbent Material	q_e (mg g ⁻¹)
Ahmad et al. [81]	Activated Alumine	130
		100
	Alum	182
		91.5
	Brick dust	115.2 100.0
Ali, Alothman, and Sanagi [28]	Iron nanoparticle impregnated via green technology	1.44
Azari et al. [17]	Iron and silver oxide nanoadsorbent	20
		21.54
		24.27
		26.3
		28.69
Babeivelri and Khodadouat [52]	Crystalline titanium dioxide powder	0.153
Barathi, Kumar, and Rajash [23]	Zirconium	1.0
	Zirconium impregnated with cellulose matrix	1.94
Basu et al. [9]	Alumina impregnated in alginate granules	3.51
		7.09
Cai et al. [51]	Hydrated aluminum oxide and anionic polyacrylamide supported on tea residue	7.43
		13.55
		18.68
Chai et al. [83]	Composite based on sulfate-doped Fe_2O_4/Al_2O_3 nanoparticles	10
		20
		34
Chen et al. [95]	Glass granules coated with iron, aluminum, and cerium-based nanoadsorbent	2.77
Cheng et al. [18]	Activated alumina	6.4
	Activated alumina modified with lanthanum nitrate	7.01
Dong and Wang [40]	Lanthanum-loaded magnetic cationic hydrogel	60.9
Gao et al. [71]	Mg, Al, and LDHs immobilized on magnetic alginate nanoflakes	32.4
He et al. [19]	hydroxyapatite nanowires	2.98
		6.28

Table 7. Cont.

Reference	Adsorbent Material	q_e (mg g ⁻¹)
Hu et al. [90]	Fe ₃ O ₄ /chitosan/Al(OH) ₃ magnetic sphere	10.45
		22.37
		74.48
Jia et al. [41]	Composite adsorbent based on bayerite and ferrhydrite	9.229
		18.17
		28.05
Jin et al. [13]	Porous MgO nanoplates	185.5
Kang et al. [56]	3 different cerium oxide morphologies—CeO ₂ (nanocubes, octahedral, nanorods)	69.3
Li et al. [44]	Activated carbon electrode charged with Ti(OH) ₄	10.4–16.5
Lin, Liu, and Cheng [58]	Composite UiO-66-NH ₂	1.55–22.8
Mahapatra, Mishra, and Hota [22]	Alumina nanofiber synthesized by the electrospinning method	1.2
Mukherjee et al. [61]	Algae biomass Nostoc sp. without treatment treated with Ca ⁺	0.6556
Rafique et al. [87]	Immobilized modified activated alumina	0.48
Raghava and Kumar [12]	Pectin composite doped with Fe-Al-Ni	7.84
	Alginate composite doped with Fe-Al-Ni	7.2
Tang and Zhang [89]	Fe (III) and Ce(IV) bimetallic oxide composite	18.99
Xiang et al. [96]	Adsorbent based on calcium, aluminum, and lanthanum nitrate	8.60
		15.24
		20.85
Xu et al. [65]	Mesoporous alumina	10.4
		77.12
		128.22
		213.84
		297.70
Yu et al. [97]	ZrO ₂ mesoporous fiber	77.12
		128.22
		213.84
		297.70
		297.70

Upon reviewing the results, a significant variation in adsorption capacity among different materials is noticeable. For instance, ZrO₂ mesoporous fiber, as reported by Yu et al. [75], demonstrates a remarkable adsorption capacity with q_e values reaching 297.70 mg g⁻¹. Some materials, such as porous MgO nanoplates (Jin et al. [13]), also exhibit impressive adsorption capacities, reaching 185.5 mg g⁻¹. On the other hand, certain adsorbents like crystalline titanium dioxide powder (Babeivelri and Khodadouat [56]) show more modest capacities, with a q_e value of 0.153 mg g⁻¹.

The variability in q_e values underscores the profound impact of the specific nature and characteristics of each adsorbent on the efficacy of fluoride removal. This understanding becomes pivotal when choosing the most suitable adsorbent for a particular fluoride removal application, considering factors such as cost, availability, and the efficiency of the adsorbent material.

Generally, an increased number of adsorbent results in a greater number of active sites available for binding F⁻ ions. However, maintaining a balance is crucial, as beyond a certain mass value, excess adsorbent does not contribute to higher efficiency per unit mass, leading to a decrease in adsorption capacity. This phenomenon is attributed to the low residual concentration of F⁻, resulting in diminished utilization efficiency.

In terms of engineering implications, the careful consideration of the optimal amount of adsorbent in a fluoride removal adsorption system is critical. Balancing process effectiveness against the associated costs of increasing the amount of adsorbent is essential and requires meticulous evaluation. Several studies by Wang et al. [50], Sapna et al. [52], Li et al. [78], Grace Lee et al. [76], Lee et al. [80], and Yu et al. [79] have explored diverse adsorbents, presenting promising results in terms of adsorption capacity. Collectively,

these studies offer insights into the potential effectiveness of various adsorbents for fluoride removal, contributing to ongoing efforts to optimize engineering solutions for water treatment processes. The diversity of applied adsorbents emphasizes the array of options available, showcasing their potential to enhance engineering solutions for water treatment.

Furthermore, the selection of the adsorbent and the optimization of its quantity and the parameters are crucial factors in the success of a fluoride removal system through adsorption. Economic considerations, such as the cost of implementing the process on a large scale, also need careful consideration when designing and implementing these processes industrially. Therefore, meticulous selection of the adsorbent and optimization of operational parameters are paramount to maximize the efficiency of fluoride removal through adsorption, ensuring the economic sustainability of the process.

6. Conclusions

This paper offers a comprehensive synthesis of a review study on fluoride remediation through adsorption, shedding light on key parameters influencing the efficacy of this process. Despite the use of diverse adsorbents, persistent challenges such as low adsorption capacity, slow adsorption rates, limited pH range, and high costs emphasize the need for a nuanced exploration of these parameters. This review underscores the pivotal role of adsorption as the primary method for fluoride removal, underscoring the importance of a comprehensive understanding for effective implementation on a broader scale. From an engineering standpoint, the imperative of addressing these challenges to optimize fluoride removal methodologies is highlighted. Furthermore, the limited pH range shows the importance of designing systems adaptable to varying water conditions, ensuring consistent fluoride removal. As technology progresses, the engineering community can explore innovative solutions to overcome these drawbacks, potentially leading to the development of more efficient adsorbents or improved processes. In conclusion, the discussion not only accentuates the current challenges in fluoride removal but also underscores the opportunity for engineers to innovate and optimize adsorption processes. A deeper understanding of the involved parameters is pivotal for refining existing methodologies and designing robust, cost-effective systems for large-scale fluoride removal, ultimately contributing to advancements in water treatment.

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